

RAINFALL PATTERN ANALYSIS USING CONTINUOUS PROBABILITY DISTRIBUTIONS

1. INTRODUCTION

Rainfall is a continuous environmental variable that varies substantially across time and location. Continuous probability distributions help describe rainfall behavior mathematically and provide useful information for water-resource planning, agriculture, flood-risk assessment and climate-related analysis.

PROBLEM STATEMENT

Rainfall observations are highly variable and positively skewed. A statistical model is required to represent the observed rainfall pattern and identify a suitable continuous distribution.

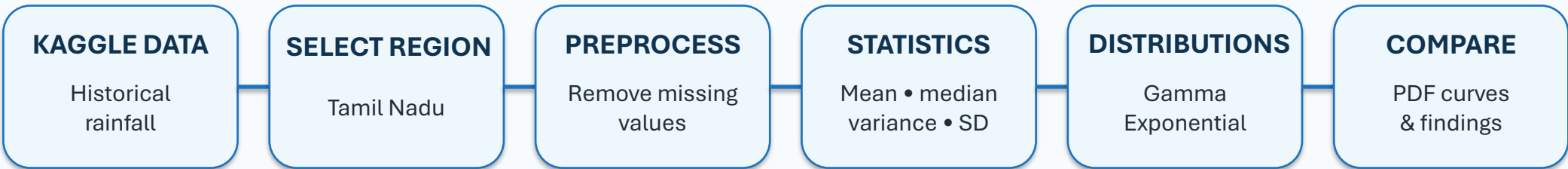
OBJECTIVES

- Analyze historical rainfall observations
- Calculate descriptive statistics
- Fit Gamma and Exponential distributions
- Compare distribution behavior using fitted curves

2. DATASET & STUDY DETAILS

DATASET	REGION	VARIABLE	OBSERVATIONS	SOURCE
Rainfall in India 1901–2015	Tamil Nadu	Monthly rainfall (mm)	1,380	Kaggle historical rainfall dataset

3. ARCHITECTURE / METHODOLOGY



4. CONTINUOUS PROBABILITY DISTRIBUTIONS

GAMMA DISTRIBUTION	EXPONENTIAL DISTRIBUTION	MODEL INTERPRETATION
Shape (α) = 0.9795 Scale (β) = 80.6435 Suitable for positive, skewed rainfall data.	Lambda (λ) = 0.0127 A simple positive-valued model for rainfall intensity.	The fitted curves closely follow the decreasing pattern of the observed rainfall distribution, showing strong right-skewness.

5. RESULTS

OBSERVATIONS	MEAN	MEDIAN	STD. DEV.	MAXIMUM
1,380	78.64 mm	59.2 mm	70.41 mm	436.1 mm

STATISTICAL RESULTS

Variance: 4957.95
Minimum: 0.0 mm
Maximum: 436.1 mm
Mean: 78.64 mm
Median: 59.2 mm
Standard deviation: 70.41 mm

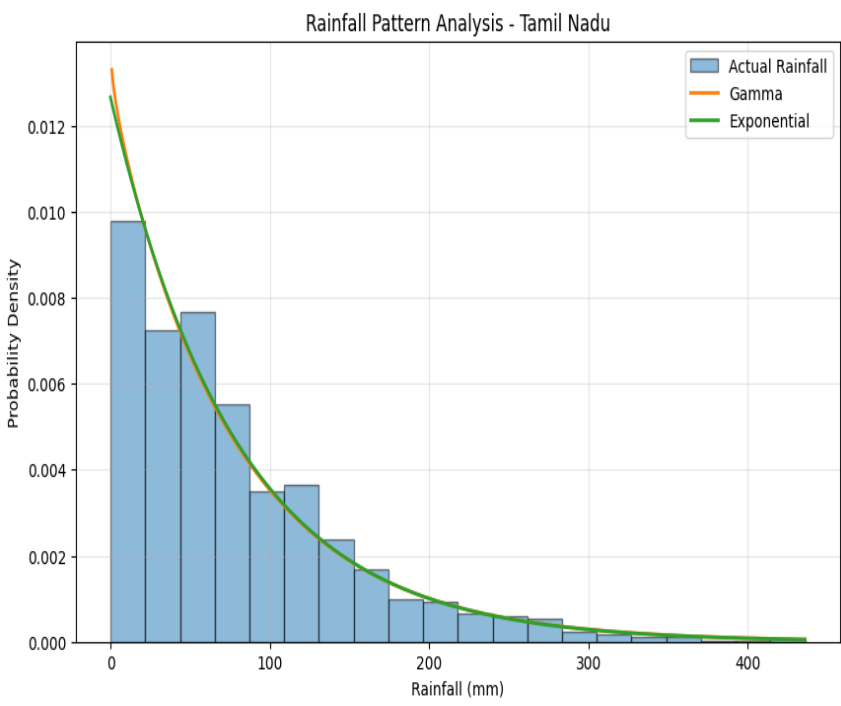
FITTED PARAMETERS

Gamma shape (α): 0.9795
Gamma scale (β): 80.6435

Exponential λ : 0.0127

Both models capture the positive and right-skewed rainfall pattern.

Observed Rainfall Distribution



6. CONCLUSION

The analysis of 1,380 historical Tamil Nadu rainfall observations shows a strongly right-skewed rainfall pattern. The mean rainfall (78.64 mm) is higher than the median (59.2 mm), indicating the influence of larger rainfall events. Gamma ($\alpha=0.9795$, $\beta=80.6435$) and Exponential ($\lambda=0.0127$) models provide continuous probabilistic representations of the observed rainfall behavior.